Polyhedra: Geometrically and Abstractly

1- and 2-Orbit Polytopes

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PART 1

Over the ages

1.0 EARLY EXPLORATIONS

1.1 In art & play

1.2 In earnest

2.0 MODERN DEVELOPMENTS

2.1 Schläfli

2.2 Petrie, Coxeter

2.3 Grünbaum, Dress

2.4 Schulte

Carved stone balls



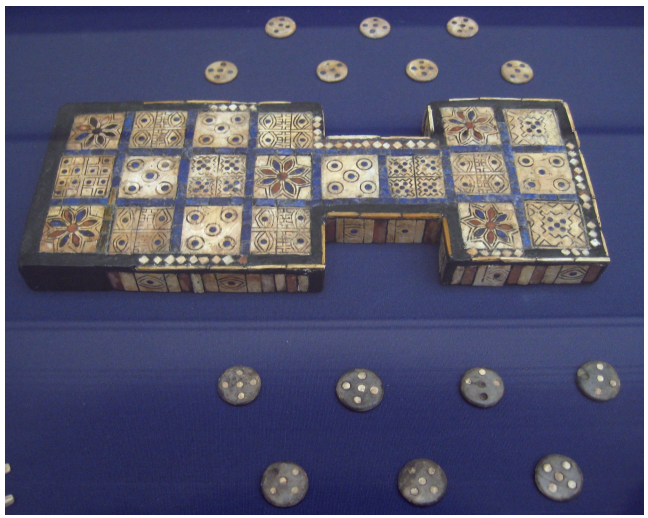


The Towie ball (engraving, 1897)

- Neolithic Scotland, c. 3200–2500 BCE; over 400 known.
- Probably not the first depiction of all five Platonic solids [1].

Left: Andy Dingley, CC BY-SA 4.0. Right: J. Evans (1897), public domain. Via Wikimedia Commons.

Dice and fairness



Royal Game of Ur, c. 2600 BCE



Icosahedral die, Ptolemaic–Roman Egypt

- **Recall:** $G(P) \stackrel{\text{def}}{=} \{\text{isometries } g\colon P \to P \mid g(P) = P\}$.
- P is fair when $G(P)$ is transitive on the planar faces.

Left: BabelStone, CC0. Right: The Metropolitan Museum of Art, 10.130.1158, CC0.

Etruscan dodecahedron

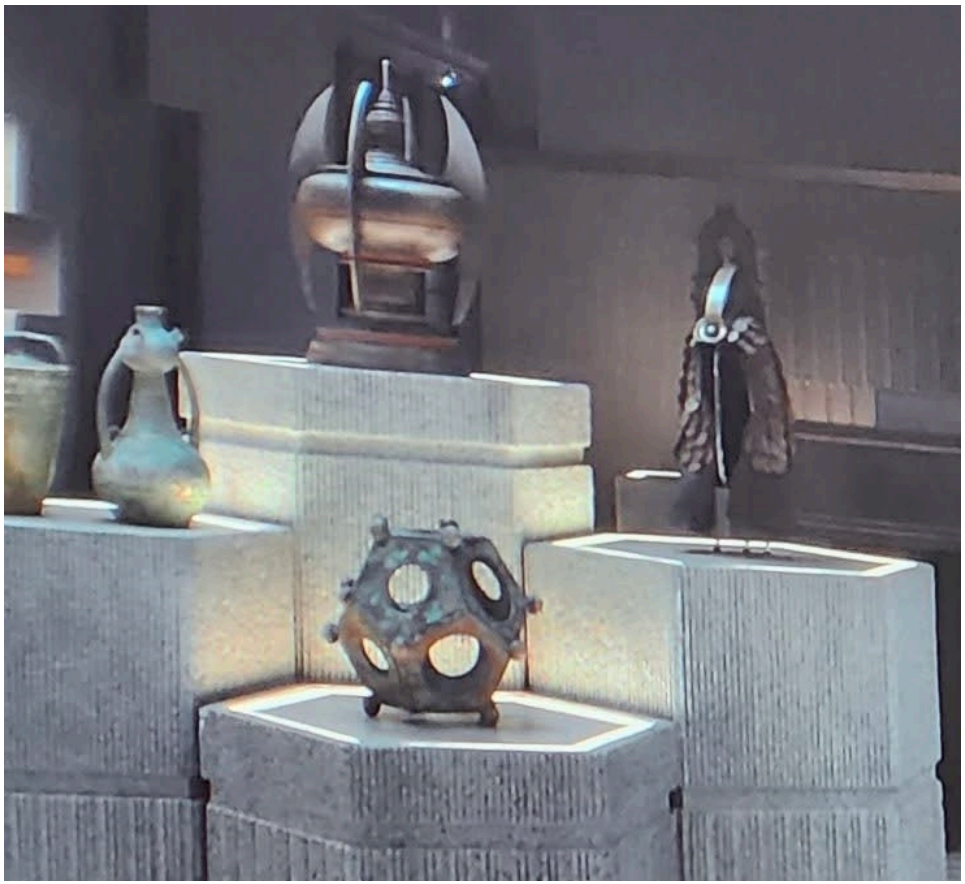




- Monte Loffa, near Verona, c. 500 BCE: a soapstone dodecahedron, likely a toy or game piece [2].
- Gallo-Roman bronze dodecahedra (2nd–4th c. CE): over 100 found across Europe. Nobody knows what they were for.

Left: Gallo-Romeins Museum Tongeren, public domain.
Right: Kleon3, CC BY-SA 4.0. Via Wikimedia Commons.

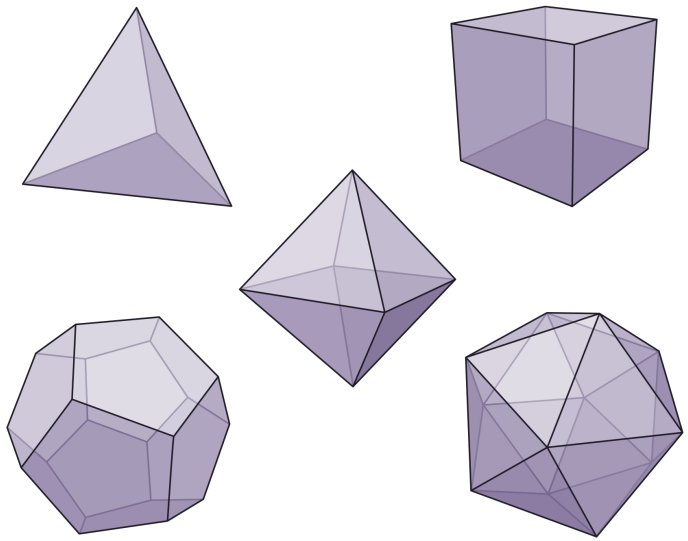
Etruscan dodecahedron



Andor (2022), Luthen Rael's gallery

Still from *Andor* (Lucasfilm / Disney+, 2022).

Platonic solids



- Plato, *Timaeus* (c. 360 BCE): fire, earth, air, water, and the cosmos.
- Theaetetus: there are exactly five.
- Euclid's *Elements*, Book XIII: the first complete catalogue.

- Faces are regular p-gons, q at each vertex. Give them a *Schläfli symbol* $\{p, q\}$.

tetrahedron	cube	octahedron	dodecahedron	icosahedron
$\{3, 3\}$	$\{4, 3\}$	$\{3, 4\}$	$\{5, 3\}$	$\{3, 5\}$

Image: Drummyfish, CC0, via Wikimedia Commons.

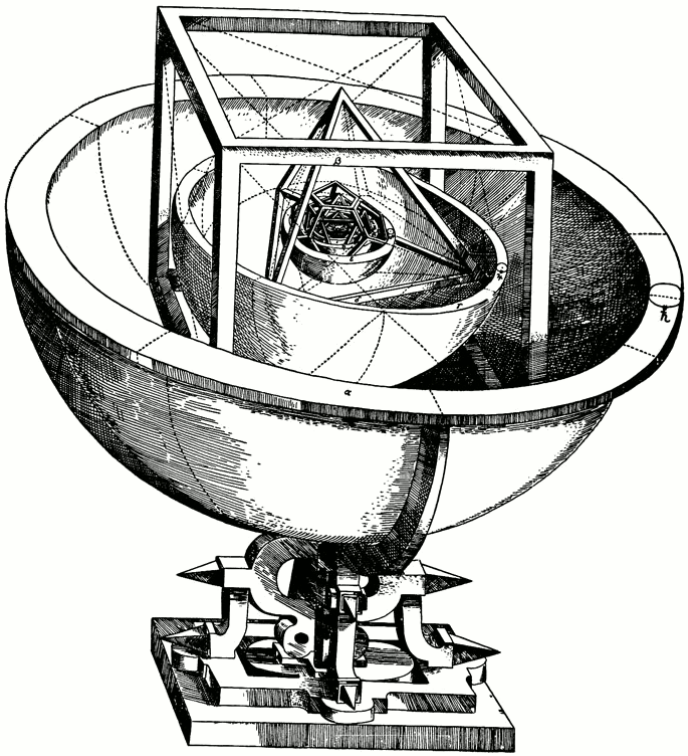
Why only five?

At a vertex... q regular p-gons meet.

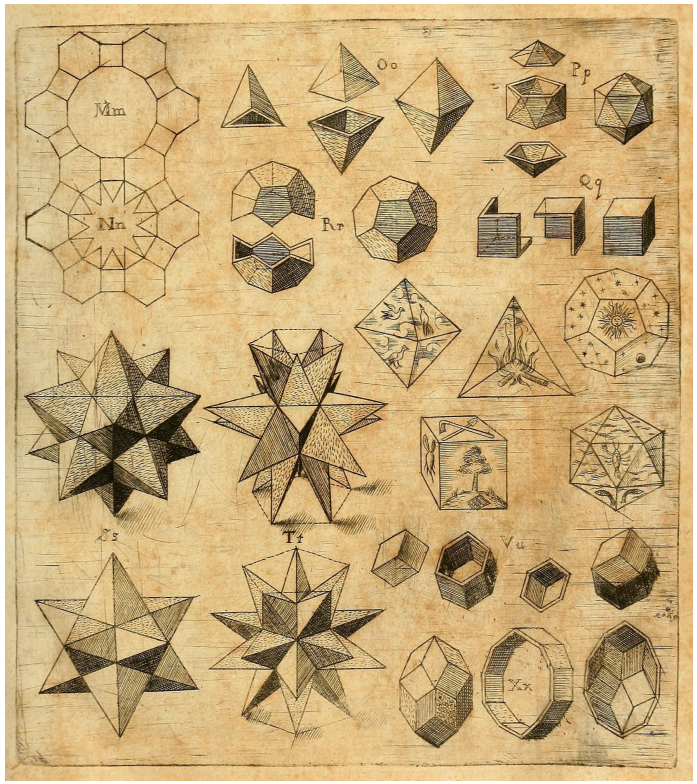
$$q\left(1-\frac{2}{p}\right)\pi < 2\pi \quad \Longleftrightarrow \quad (p-2)(q-2) < 4.$$

$p\backslash q$	3	4	5	6
3	$\{3,3\}$	$\{3,4\}$	$\{3,5\}$	$\{3,6\}$
4	$\{4,3\}$	$\{4,4\}$	$\{4,5\}$	$\{4,6\}$
5	$\{5,3\}$	$\{5,4\}$	$\{5,5\}$	$\{5,6\}$
6	$\{6,3\}$	$\{6,4\}$	$\{6,5\}$	$\{6,6\}$

Kepler's cosmos [3]



Mysterium Cosmographicum (1596)

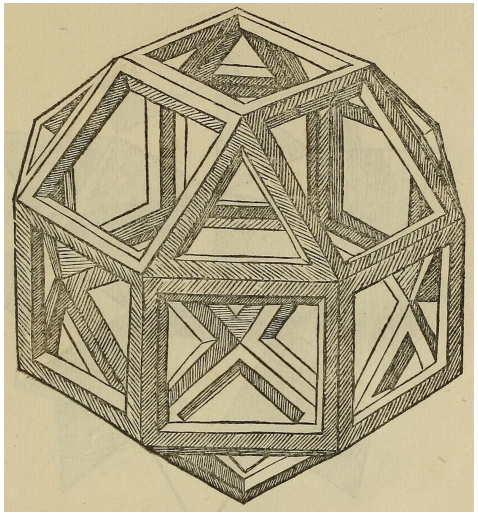


Harmonices Mundi (1619)

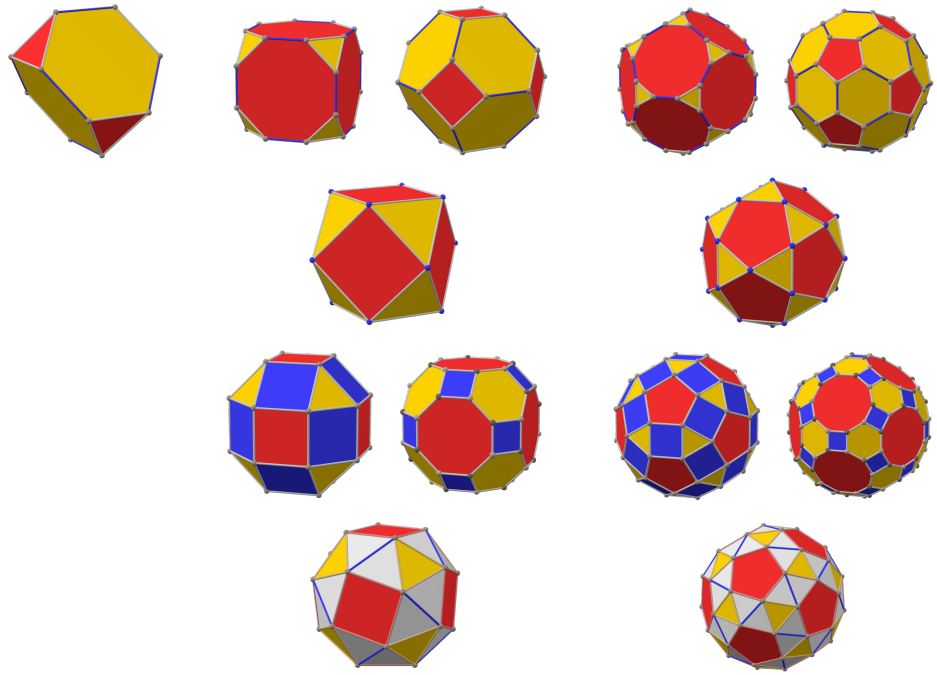
- Orbits of the six (known) heavenly bodies match the five solids (+ a sphere).
- Whimsical!

Both images public domain, via Wikimedia Commons.

Archimedean solids



Leonardo's
rhombicuboctahedron (1509)

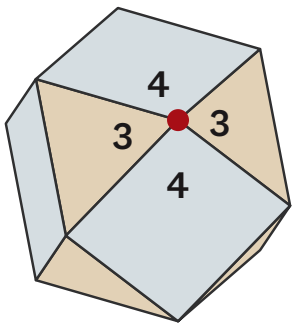


The 13 Archimedean solids

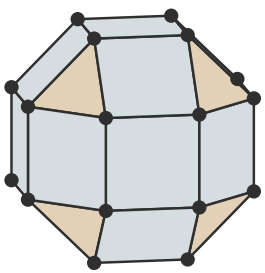
- Faces are regular polygons of two or more kinds.
- $G(P)$ is transitive on vertices.
- Archimedes' 13, known only through Pappus; completed by Kepler [4], who added the prisms and antiprisms.

Left: public domain. Right: Tilman Piesk, CC BY 4.0. Via Wikimedia Commons.

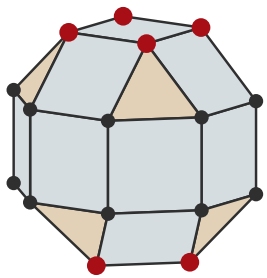
Naming by vertex configuration



- The *cuboctahedron* is 3.4.3.4.



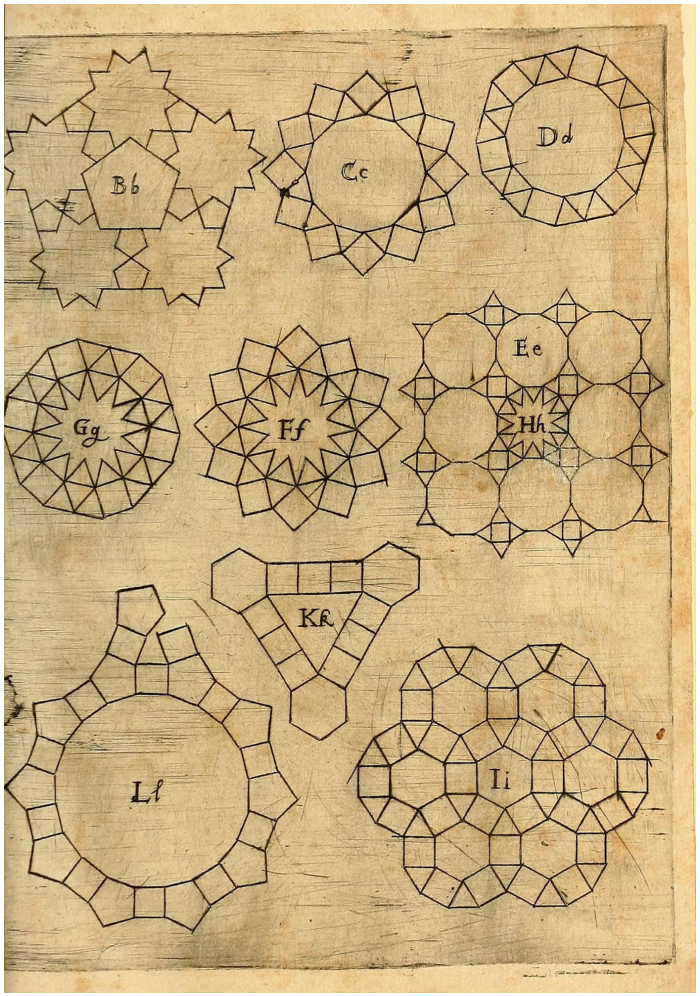
rhombicuboctahedron



pseudo-rhombicuboctahedron

- **Warning:** the same faces at every vertex is not enough [5].

Uniform tilings

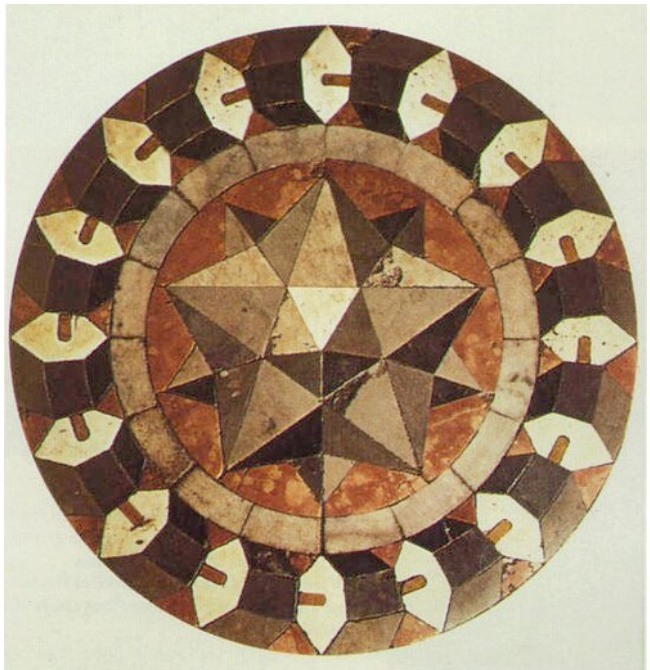


- Kepler also found the 11 tilings of the plane by regular polygons with all vertices alike [4].

Transitive?	*V*	*E*	*F*
Platonic	✓	✓	✓
Archimedean	✓		
cuboctahedron	✓	✓	
Catalan (duals)			✓

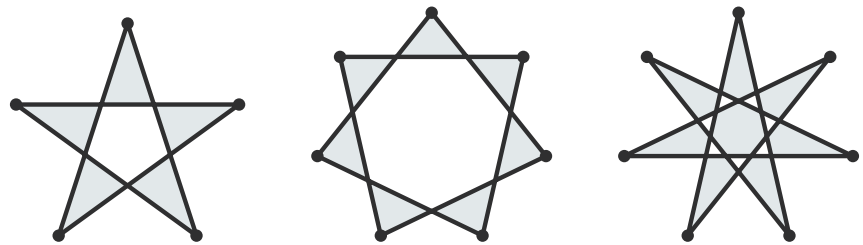
Image: Kepler, *Harmonices Mundi* (1619), public domain.

Star polygons



Attr. Paolo Uccello, St Mark's Basilica, Venice (c. 1430)

- *Star polygon* $\{n/k\}$: join every kth of n evenly-spaced points.

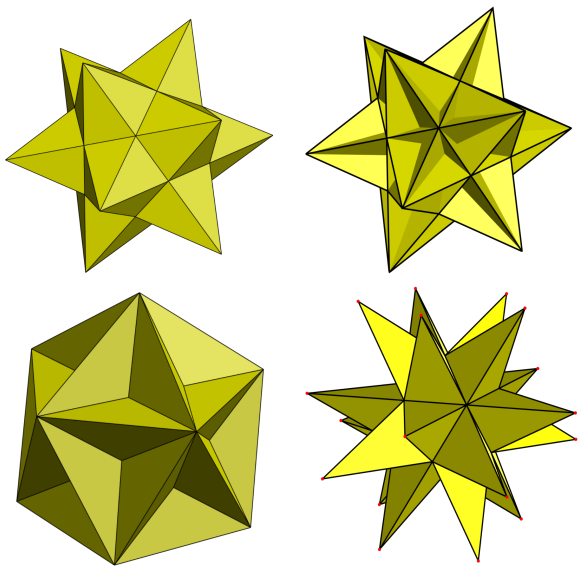


$\{5/2\}$ $\{7/2\}$ $\{7/3\}$

- First assumption dropped: faces need not be convex; they may cross themselves.

Mosaic photo: public domain, via Wikimedia Commons.

Star polyhedra

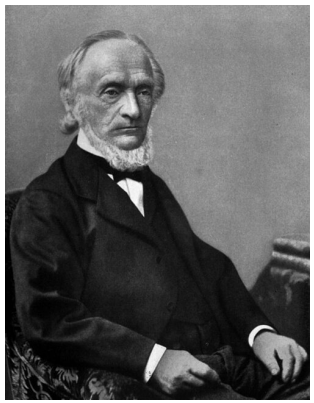


$\{5/2, 5\}$	small stellated dodec.	Kepler [4]
$\{5/2, 3\}$	great stellated dodec.	—
$\{5, 5/2\}$	great dodecahedron	Poinsot [6]
$\{3, 5/2\}$	great icosahedron	—

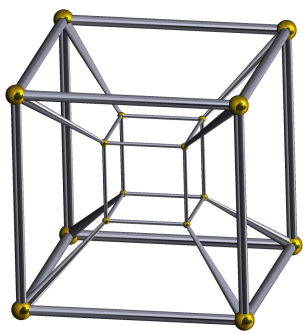
- Cauchy [7] showed there are no others.
- Faces may be stars, or meet in a star pattern at a vertex.

Image: Dedhert.Jr, CC BY 4.0, via Wikimedia Commons.

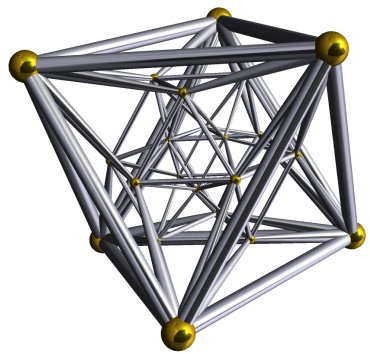
In 4-space



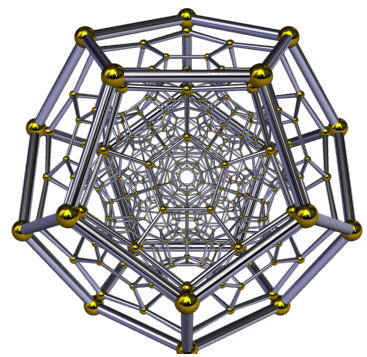
- Schläfli, *Theorie der vielfachen Kontinuität* (1850–52, published 1901) [8].
- Six convex regular 4-polytopes: $\{3,3,3\}$, $\{4,3,3\}$, $\{3,3,4\}$, $\{3,4,3\}$, $\{5,3,3\}$, $\{3,3,5\}$.
- In every dimension ≥ 5, only three. (The 24-cell has no analogue in any other dimension.)



$\{4,3,3\}$ 4-cube



$\{3,4,3\}$ 24-cell



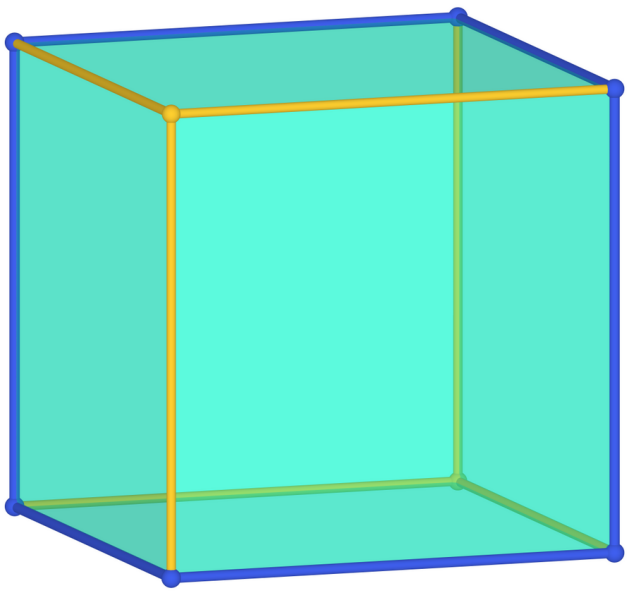
$\{5,3,3\}$ 120-cell

Schlegel diagrams: Image created with Robert Webb's Stella software, CC BY-SA 3.0. Portrait: public domain.

What counts as “genuine”?

	V	E	F	$V-E+F$
$\{5/2,5\}$	12	30	12	-6
$\{5,5/2\}$	12	30	12	-6
$\{5/2,3\}$	20	30	12	2
$\{3,5/2\}$	12	30	20	2

Petrie polygons

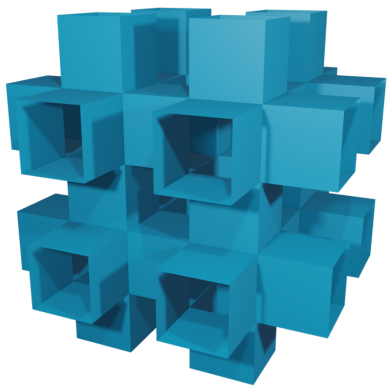


- A *Petrie polygon* is a zigzag path of edges.
- For the cube, Petrie polygons are skew hexagons.
- Named by Coxeter after his friend John Flinders Petrie [9].

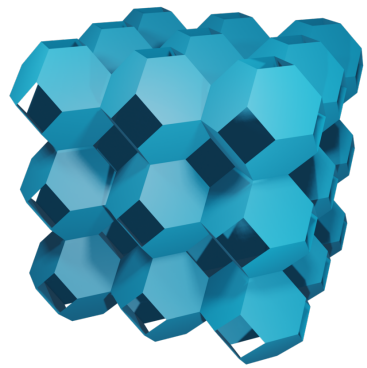
Image: Carsten Steger, CC BY-SA 4.0, via Wikimedia Commons.

Regular skew polyhedra

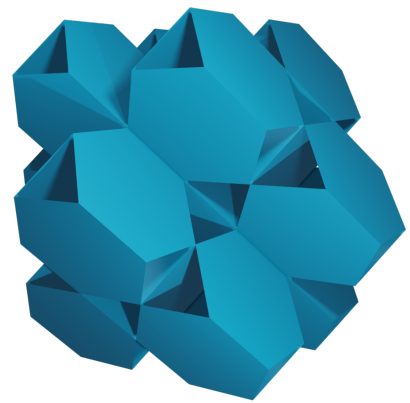
- Petrie and Coxeter (1926) let the *vertex-figures* be skew and found three infinite regular polyhedra [10]:



$\{4, 6 \mid 4\}$



$\{6, 4 \mid 4\}$



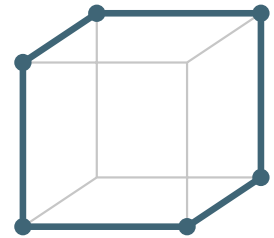
$\{6, 6 \mid 3\}$

- Second assumption dropped: finiteness.

Images: Sycamore916, CC BY-SA 4.0, via Wikimedia Commons.

First operations

- *Duality* δ: swap vertices and faces, so $\{p, q\} \mapsto \{q, p\}$.
- *Petrial* π: keep the vertices and edges, and use the Petrie polygons as the new faces.



a face of the cube's Petrial

Example

Cube $\{4, 3\} \xrightarrow{\pi} \{6, 3\}_4$: 8 vertices, 12 edges, and 4 skew hexagons.

- $\pi^2 = \delta^2 = (\pi\delta)^3 = 1$, so $\langle \pi, \delta \rangle \cong S_3$.
- Each P has up to six relatives:

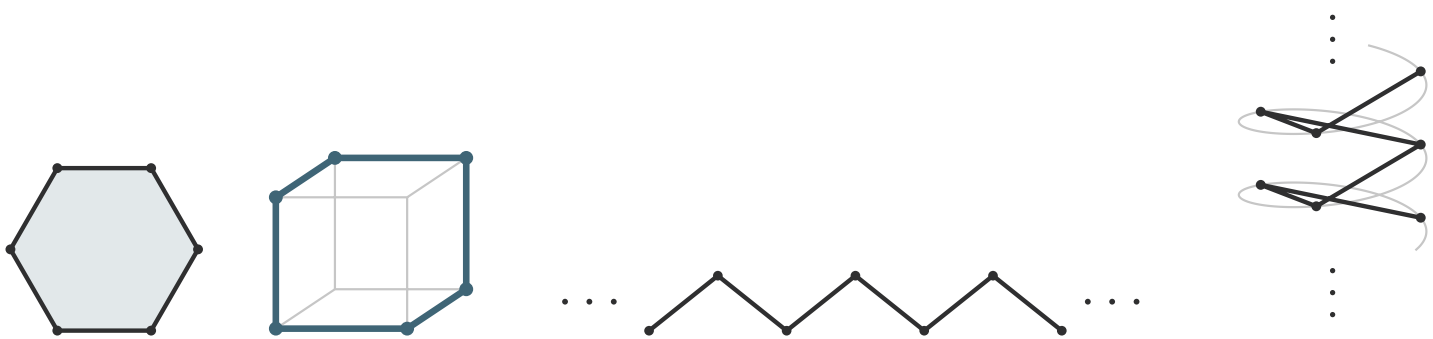
$$P, \quad P^{\delta}, \quad P^{\pi}, \quad P^{\pi\delta}, \quad P^{\delta\pi}, \quad P^{\pi\delta\pi}.$$

Towards abstraction



Grünbaum

- Grünbaum, *Regular polyhedra — old and new* (1977) [11]: a polygon is any sequence of points joined in order, either finite and closed, or infinite in both directions (an *apeirogon*).
- A *polyhedron* is a family of such polygons glued edge to edge. Say P is *regular* if $G(P)$ is transitive on flags.



Portrait: Konrad Jacobs, © MFO, CC BY-SA 2.0 de.

Enumeration of regular polyhedra



Dress

18	*finite*: 5 Platonic, 4 Kepler–Poinsot, 9 Petrials
6	*planar*: 3 tilings and their Petrials
12	*blended*: a planar and a linear part
12	*pure*: including Petrie–Coxeter's 3
48	

- Grünbaum (1977) found 47.
- Dress found the 48th (1981) and proved the list complete (1985) [12, 13].
- McMullen and Schulte (1997) gave a short proof [14].

Portrait: Konrad Jacobs, © MFO, CC BY-SA 2.0 de.

All 48, on video



A popular (3.2M views) and digestible video on the 48 regular polyhedra can be found here.

Video frame: jan Misali, via YouTube.

The 18 finite regular polyhedra

TETRAHEDRAL

$$\{3,3\} \overset{\pi}{\longleftrightarrow} \{4,3\}_3$$

OCTAHEDRAL

$$\{6,4\}_3 \overset{\pi}{\longleftrightarrow} \{3,4\} \overset{\delta}{\longleftrightarrow} \{4,3\} \overset{\pi}{\longleftrightarrow} \{6,3\}_4$$

ICOSAHEDRAL

$$\begin{array}{ccccccc}
\{10,5\} & \overset{\pi}{\longleftrightarrow} & \{3,5\} & \overset{\delta}{\longleftrightarrow} & \{5,3\} & \overset{\pi}{\longleftrightarrow} & \{10,3\} \\
\updownarrow \varphi_2 & & \updownarrow \varphi_2 & & & & \\
\{6,5/2\} & \overset{\pi}{\longleftrightarrow} & \{5,5/2\} & \overset{\delta}{\longleftrightarrow} & \{5/2,5\} & \overset{\pi}{\longleftrightarrow} & \{6,5\} \\
 & & & & \updownarrow \varphi_2 & & \updownarrow \varphi_2 \\
\{10/3,3\} & \overset{\pi}{\longleftrightarrow} & \{5/2,3\} & \overset{\delta}{\longleftrightarrow} & \{3,5/2\} & \overset{\pi}{\longleftrightarrow} & \{10/3,5/2\}
\end{array}$$

π: Petrial $\quad$ δ: dual $\quad$ φ_2: facetting, $(R_0, R_1, R_2) \mapsto (R_0, R_1R_2R_1, R_2)$

Diagrams after McMullen–Schulte [14].

The 6 planar and 12 blended apeirohedra

PLANAR

$$\{4,4\} \overset{\pi}{\longleftrightarrow} \{\infty,4\}_4 \qquad \{\infty,6\}_3 \overset{\pi}{\longleftrightarrow} \{3,6\} \overset{\delta}{\longleftrightarrow} \{6,3\} \overset{\pi}{\longleftrightarrow} \{\infty,3\}_6$$

BLENDED WITH A SEGMENT

$$\{4,4\} \# \{\} \overset{\pi}{\longleftrightarrow} \{\infty,4\}_4 \# \{\}$$

$$\{3,6\} \# \{\} \overset{\pi}{\longleftrightarrow} \{\infty,6\}_3 \# \{\}$$

$$\{6,3\} \# \{\} \overset{\pi}{\longleftrightarrow} \{\infty,3\}_6 \# \{\}$$

BLENDED WITH A LINE

$$\{4,4\} \# \{\infty\} \overset{\pi}{\longleftrightarrow} \{\infty,4\}_4 \# \{\infty\}$$

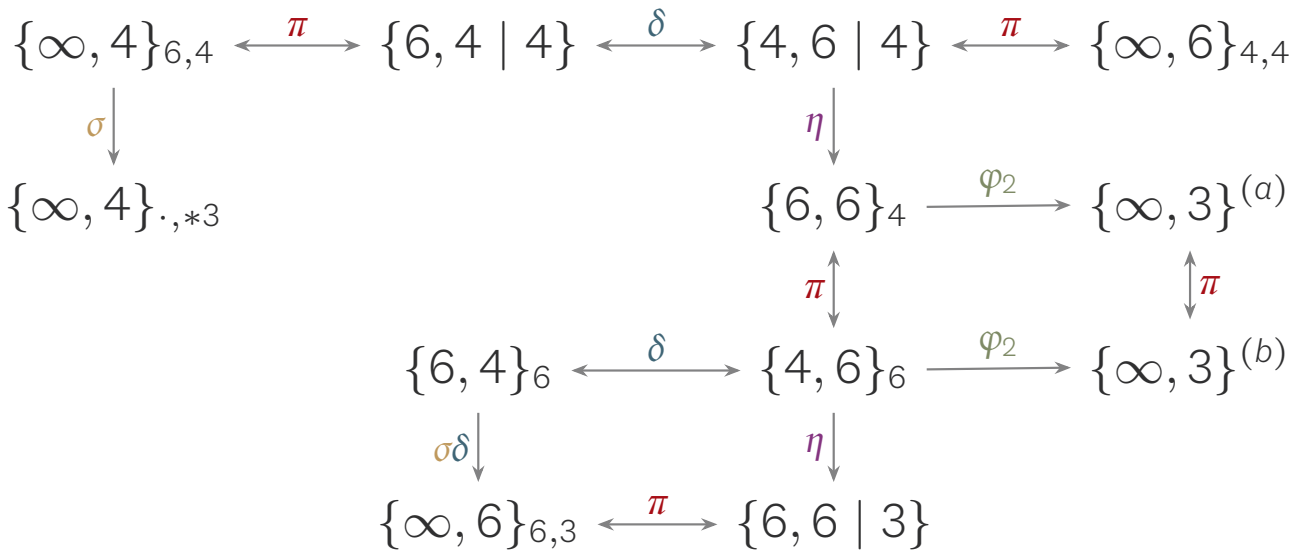
$$\{3,6\} \# \{\infty\} \overset{\pi}{\longleftrightarrow} \{\infty,6\}_3 \# \{\infty\}$$

$$\{6,3\} \# \{\infty\} \overset{\pi}{\longleftrightarrow} \{\infty,3\}_6 \# \{\infty\}$$

$P \# Q$: the *blend* of a planar P with a segment or a linear apeirogon Q.

Diagrams after McMullen–Schulte [14].

The 12 pure apeirohedra



All derived from the cubical honeycomb $\{4, 3, 4\}$; the top row contains Petrie–Coxeter's $\{6, 4 \mid 4\}$ and $\{4, 6 \mid 4\}$.

η: halving, $(R_0, R_1, R_2) \mapsto (R_0R_1R_0, R_2, R_1)$ $\quad$ $\sigma = \pi\delta\eta\pi\delta$: skewing

φ_2: facetting, as before

Diagram after McMullen–Schulte [14].

Enumeration of chiral polyhedra

- *Chiral*: the symmetry group has two orbits on flags, and adjacent flags lie in different orbits.

Schulte (2004, 2005) [15, 16] classified the chiral polyhedra in $\mathbb{E}^3$:

- None are finite, and none are planar.
- Three families with finite skew faces, of types $\{6,6\}$, $\{4,6\}$, $\{6,4\}$.
- Three families with helical faces, of types $\{\infty,3\}$ and $\{\infty,4\}$.

PART 2

Contemporary abstractions

3.0 MAPS

4.0 ABSTRACT POLYTOPES

4.1 Highly symmetric abstract polytopes

Forget the ambient space

Each step so far dropped a geometric assumption:

$$\text{convexity} \to \text{finiteness} \to \text{planar faces} \to \text{finite faces}$$

Only incidence: which vertices lie on which edges, and which edges lie on which faces.

Idea

A polyhedron *is* its combinatorics.

Maps on surfaces

A *map* is a family of polygons (possibly apeirogons) such that

(a) any two polygons meet in a common edge, a common vertex, or not at all;

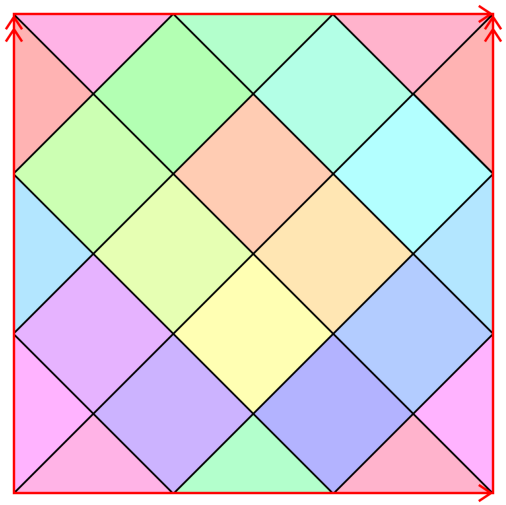
(b) each edge belongs to exactly two polygons;

(c) the polygons containing a given vertex form a single cycle of adjacent polygons (sharing a common edge);

(d) between any two polygons is a chain of adjacent polygons.

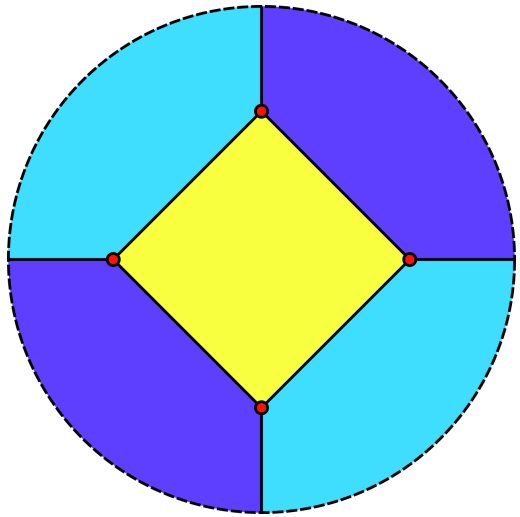
Glue the polygons along their edges and you get a surface. Its genus g is read off from $V - E + F = 2 - 2g$ (orientable case).

Examples of maps



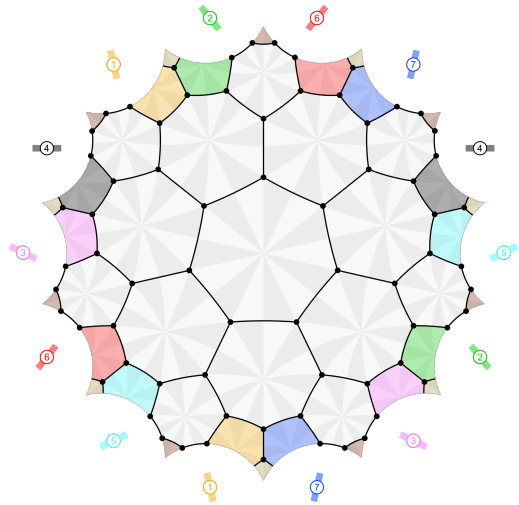
$\{4,4\}_{(3,3)}$ on the torus

glue opposite sides



hemicube

projective plane



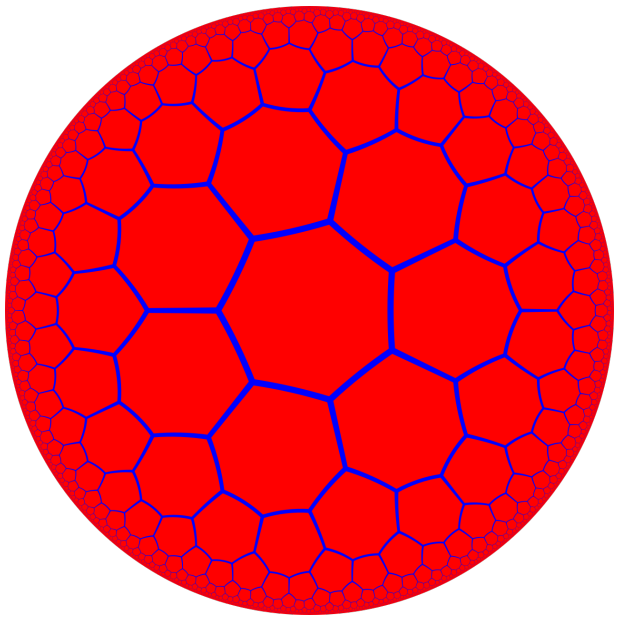
Klein's map $\{7,3\}_8$ [17]

genus 3

- The *hemicube*: identify opposite faces of the cube. 3 squares, 6 edges, 4 vertices.
- *Klein's map* is realized in a few trippy ways!

Left: Garygo golob. Middle: AquitaneHungerForce. Right: Tilman Piesk. All CC BY-SA 4.0, via Wikimedia Commons.

Resolving Schläfli's gripes



$\{7, 3\}$ in the Poincaré disk

- The great dodecahedron $\{5, 5/2\}$, as a map: 12 pentagons, 5 at each vertex.

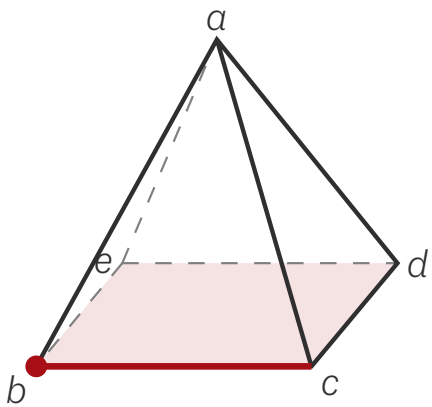
$$V - E + F = -6 = 2 - 2g \qquad \Longrightarrow \qquad g = 4.$$

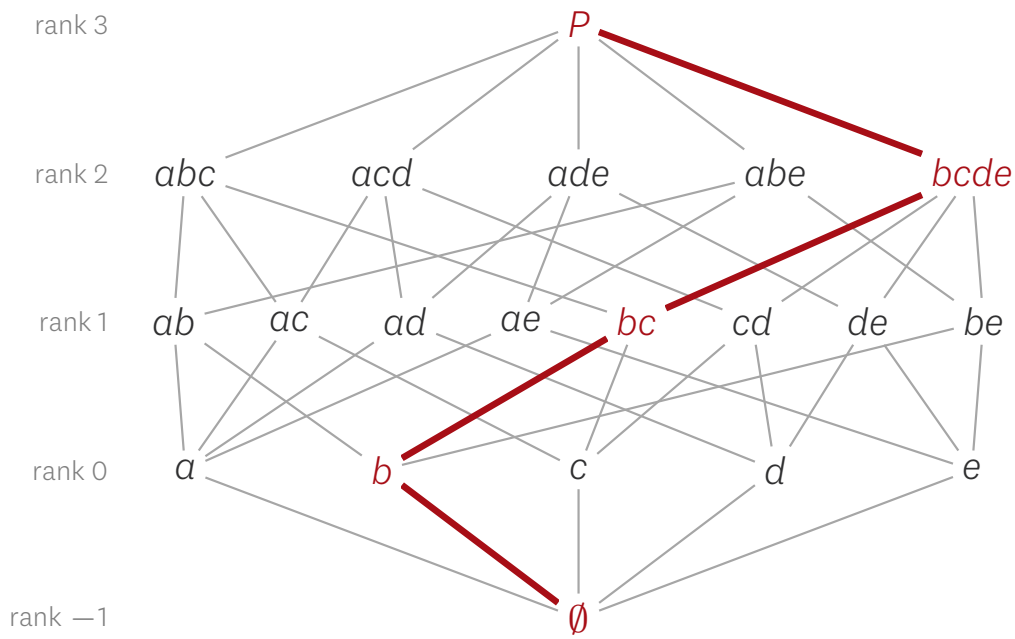
- The $\{p, q\}$ with $(p-2)(q-2) > 4$ tile the hyperbolic plane. Klein's map is a finite quotient of $\{7, 3\}$.

Image: Parcly Taxel, public domain, via Wikimedia Commons.

Faces as a poset

- Order the faces by inclusion, then add $\emptyset$ and the whole polytope P:





- A *flag* is a maximal chain, e.g. $\emptyset < b < bc < bcde < P$.

DEFINITION

Abstract polytopes

An *abstract n-polytope* is a ranked poset $\mathcal{P} = (\mathcal{P}, \preccurlyeq)$ satisfying

(P1) There is a least and greatest face.

(P2) Every flag has $n + 2$ faces.

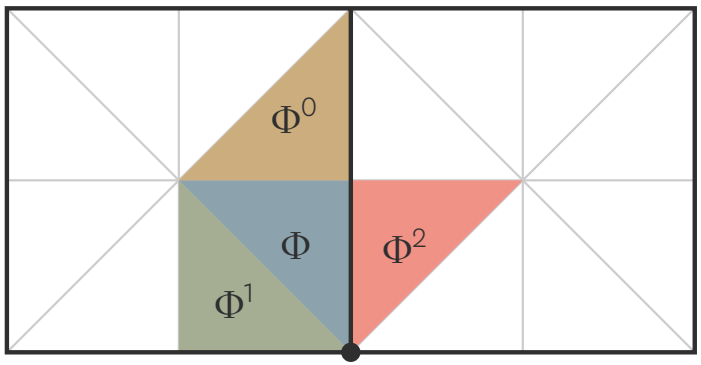
(P3) The flags are *strongly-connected*.

For an arbitrary $F \in \mathcal{P}$ with $\mathbf{rk}(F) = i$, write F_i.

(P4) For each $i = 0, \ldots, n-1$ and every $F_{i-1} \preccurlyeq F_{i+1}$, there exist exactly two F_i with

$$F_{i-1} \preccurlyeq F_i \preccurlyeq F_{i+1}.$$

Adjacent flags

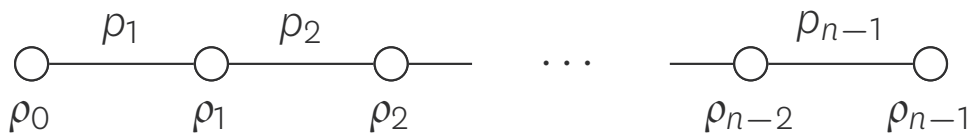


Φ^i differs from Φ in exactly the rank-i face.

- (P4) says Φ^i exists and is unique.
- (P3) says any flag can be reached from any other by such moves.

Abstract regular polytopes

- Say $\mathcal{P}$ is *regular* if $\Gamma(\mathcal{P}) \stackrel{\text{def}}{=} \mathrm{Aut}(\mathcal{P})$ acts transitively on the flags.
- There's a presentation of $\Gamma(\mathcal{P})$ as a *string diagram*:



$$\Gamma(\mathcal{P}) = \langle \rho_0, \ldots, \rho_{n-1} \rangle = [p_1, \ldots, p_{n-1}]/N$$

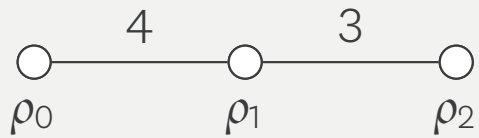
- The ρ_i's act on a *base flag* Φ via $\Phi\rho_i \stackrel{\text{def}}{=} \Phi^i$, and the subgroups $\langle \rho_i \mid i \in I \rangle$ ($I \subseteq \{0, \ldots, n-1\}$) satisfy an intersection condition.

Theorem (McMullen & Schulte [18])

Abstract regular polytopes and string C-groups are the same object.

Abstract regular polytopes

Example (The cube $\{4,3\}$)



$$\Gamma = [4,3] = \langle \rho_0, \rho_1, \rho_2 \mid \rho_i^2 = (\rho_0\rho_1)^4 = (\rho_1\rho_2)^3 = (\rho_0\rho_2)^2 = 1 \rangle$$

has order 48: exactly one automorphism per flag.

Example (The hemicube)

Identify antipodal faces:

$$\Gamma = [4,3] \,/\, \left\langle (\rho_0\rho_1\rho_2)^3 \right\rangle \cong S_4,$$

of order 24.

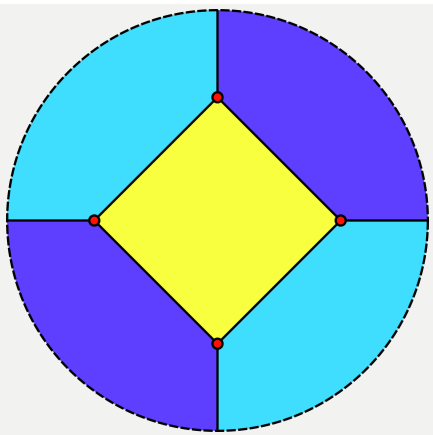


Image: AquitaneHungerForce, CC BY-SA 4.0, via Wikimedia Commons.

Chiral polytopes

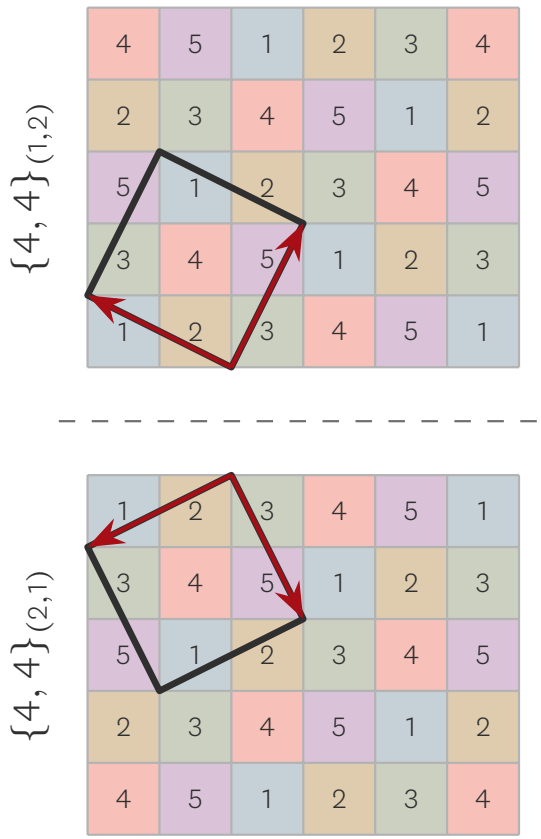
- $\mathcal{P}$ *chiral* if $\Gamma(\mathcal{P})$ induces exactly **two** orbits on the flags, where adjacent flags always lie in opposite orbits.
- There's a presentation $\Gamma(\mathcal{P}) = \langle \sigma_1, \ldots, \sigma_{n-1} \rangle$ by *abstract rotations*, but $\Gamma(\mathcal{P})$ admits no involutory *enantiomorphism* $\eta \in \mathrm{Aut}(\Gamma(\mathcal{P}))$:

$$\Gamma(\mathcal{P}_{(\Phi)}) \xleftrightarrow{\eta} \Gamma(\mathcal{P}_{(\Phi^0)})$$

- **In other words...** $\Gamma(\mathcal{P}) = [p_1, \ldots, p_{n-1}]^+ / N$ for some N which is *not* normal in the full $[p_1, \ldots, p_{n-1}]$.

EXAMPLE

Chiral polytopes



- The torus map $\{4,4\}_{(1,2)}$: glue the square tiling by the translations $(1,2)$ and $(-2,1)$.
- 5 faces, 10 edges, 5 vertices, 40 flags.
- All 20 rotations survive the gluing. No reflection does: a reflection sends $(1,2)$ to $(2,1)$, which gives the mirror image map (below).

DEFINITION

2-orbit polytopes

What if $\Gamma(\mathcal{P})$ has exactly 2 flag-orbits but is not necessarily chiral?

- Then there's some *proper* $I \subset \{0, \ldots, n-1\}$ with Φ, Φ^i in the same orbit for every $i \in I$.
- Say $\mathcal{P}$ is in *class* $\mathbf{2}_I$. **Note:** $\mathbf{2}_\emptyset$ is the class of chiral polytopes.
- So there are $2^n - 1$ classes of 2-orbit n-polytopes [19].

DEFINITION

2-orbit polytopes

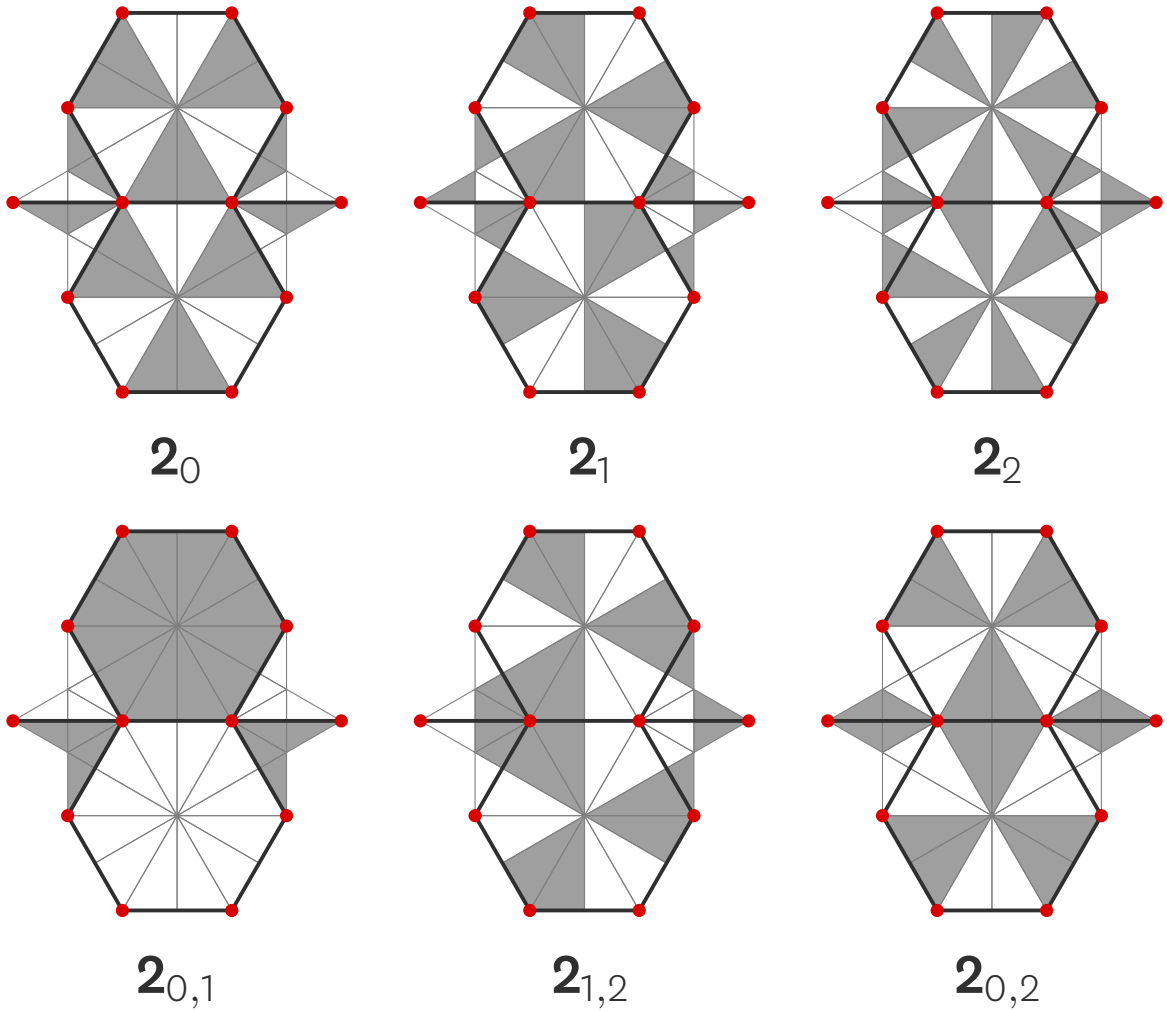
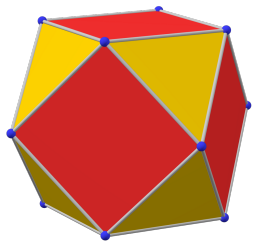
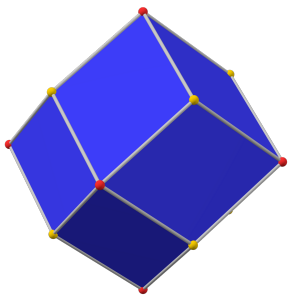


Figure 1: Local flags of non-chiral polyhedra of type $\{6, 4\}$ in $\mathbf{2}_I$. [19]

2-orbit polytopes



cuboctahedron



rhombic dodecahedron

96 flags, but only 48 symmetries.

- Reflections swap $\Phi \leftrightarrow \Phi^0$ and $\Phi \leftrightarrow \Phi^1$.
- Nothing swaps a triangle with a square, so Φ and Φ^2 lie in different orbits.

The cuboctahedron is in class $\mathbf{2}_{\{0,1\}}$.

Its dual, the rhombic dodecahedron, is in $\mathbf{2}_{\{1,2\}}$.

Images: Tilman Piesk, CC BY 4.0, via Wikimedia Commons.

2-orbit polytopes

Example (Polyhedra of type $\{2p_1, 2p_2\}$ in $2_{0,2}$)

Looking for groups G and generators $\rho_0, \rho_2, \alpha_{101}, \alpha_{121}$ satisfying

$$\begin{aligned}\rho_0^2 = \rho_2^2 = \alpha_{101}^2 = \alpha_{121}^2 &= (\rho_0\rho_2)^2 \\ &= (\alpha_{101}\alpha_{121})^2 = (\rho_0\alpha_{101})^{p_1} \\ &= (\rho_2\alpha_{121})^{p_2} = 1,\end{aligned}$$

as well as five intersection conditions

$$\begin{aligned}\langle \rho_0, \alpha_{101}\rangle \cap \langle \rho_2, \alpha_{121}\rangle &= 1, \\ \langle \rho_0, \alpha_{101}\rangle \cap \langle \rho_0, \rho_2\rangle &= \langle \rho_0\rangle, \\ \langle \rho_2, \alpha_{121}\rangle \cap \langle \rho_0, \rho_2\rangle &= \langle \rho_2\rangle, \\ \langle \rho_0, \alpha_{101}\rangle \cap \langle \alpha_{101}, \alpha_{121}\rangle &= \langle \alpha_{101}\rangle, \\ \langle \rho_2, \alpha_{121}\rangle \cap \langle \alpha_{101}, \alpha_{121}\rangle &= \langle \alpha_{121}\rangle.\end{aligned}$$

PART 3

My research

Where my research lives

Recall:

$$\mathrm{AGL}(1,q) \stackrel{\text{def}}{=} \left\{ x \mapsto ax + b \;\middle|\; a \in \mathbb{F}_q^* \text{ and } b \in \mathbb{F}_q \right\} \cong \mathbb{F}_q \rtimes \mathbb{F}_q^*.$$

It turns out these give chiral polyhedra, and their Petrials give 2-orbit polyhedra [20–22].

Questions

- For which G and $\mathbb{Z}[G]$-modules V is $\Gamma \leq V \rtimes G$ the group of a 1- or 2-orbit n-polytope?
- It seems such $V \rtimes G$ is the simplest way to construct them from scratch. ***Why?***
- They are a majority of chiral polyhedra of order $2^n, 2^n p$. ***Why?***

THANKS!

Questions?

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